

GearDesign Report

Cylindrical gear(pair)

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1. Summary

1.1. Gear geometry

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Normal module (mm)	m_n	2.5000	
Pressure angle at normal section (deg)	α_n	20.0000	
Helix angle at reference circle (deg)	β	23.1000	
Manual normal backlash (mm)	j_{bn}	0.0000	
Number of teeth	z	20.0000	44.0000
Facewidth (mm)	b	40.0000	40.0000
Profile shift coefficient	x	0.0000	0.0000
Quality (ISO)	Q	6.0000	6.0000
Center distance (mm)	a	86.9730	
Tooth thickness tolerance	-	DIN 3967 cd25	DIN 3967 cd25
Gear material	-	DIN 18CrNiMo7 ($\sigma_{Hlim} = 430$ Mpa)	DIN 18CrNiMo7 ($\sigma_{Hlim} = 430$ Mpa)
Reference diameter (mm)	d	54.3580	119.5880
Tip diameter (mm)	d_a	59.358 (59.358 / 59.258)	124.588 (124.588 / 124.488)
Root diameter (mm)	d_f	48.108 (47.916 / 47.806)	113.338 (113.146 / 113.036)
Number of teeth spanned	k	3.0000	7.0000
Base tangent length (no backlash) (mm)	W_k	19.339 (19.273 / 19.236)	49.926 (49.861 / 49.823)
Effective Diameter of ball/pin (mm)	D_M	4.5000	4.2500
Diametral measurement over two balls (mm)	M_{dk}	60.987 (60.828 / 60.736)	125.412 (125.233 / 125.130)

1.2. Total safety factor

	Gear1 [Left]	Gear1 [Right]	Gear2 [Left]	Gear2 [Right]	Required S.F
Contact S.F	1.1353		1.1631		1.0000
Bending S.F	2.7795		2.9428		1.4000
Scuffing S.F (Flash Temp.)	2.1823				2.0000
Micropitting S.F	0.7144				1.0000

1.3. Total damage & life

	Gear1 [Left]	Gear1 [Right]	Gear2 [Left]	Gear2 [Right]
Contact Damage [%]	1.6000	0.0000	0.7000	0.0000
Bending Damage [%]	0.0000	0.0000	0.0000	0.0000
Contact Life [hr]	∞	∞	∞	∞
Bending Life [hr]	∞	∞	∞	∞
Total Req. Life [hr]	10000.0000			

1.4. Others

	Gear1 - Gear2
Transverse Contact Ratio	1.4157
Overlap Ratio	1.9982
Total Contact Ratio	3.4138
Worst Pitch Line Velocity [m/s]	28.4620
Total Efficiency [%]	98.7909
Worst Mesh Loss [W]	795.6138
Worst Churning Loss [W]	1018.0740
Worst Peak-to-Peak T.E. [μm]	0.2488
Gear Total Mass [kg]	1.5960
Hunting tooth	Patial

1.5. Summary - Load cases

	Gear	Time [Hr]	Temperaure [°C]	Power [kW]	Speed [r/min]	Torque [N.m]	Contact SF (Left)	Bending SF (Left)	Scuffing SF (Left)	Contact SF (Right)	Bending SF (Right)	Scuffing SF (Right)	K_Hβ (Left)	K_Hβ (Right)	Mesh Loss (Left) [kW]	Mesh Loss (Right) [kW]	Churning Loss [kW]	PPTe (Left) [μm]	PPTe (Right) [μm]
LC	Gear1	10000.0000	80.0000	150.0000	10000.0000	143.2390	1.1353	2.7795	2.1823				1.3000		0.7956		0.1363	0.2488	
1	Gear2			-150.0000	-4545.4550	315.1270	1.1631	2.9428									0.8817		

2. Detail geometry

2.1. Input gear geometry

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Normal module (mm)	m_n	2.5000	
Pressure angle at normal section (deg)	α_n	20.0000	
Helix angle at reference circle (deg)	β	23.1000	
Manual normal backlash (mm)	j_{bn}	0.0000	
Number of teeth	z	20.0000	44.0000
Facewidth (mm)	b	40.0000	40.0000
Profile shift coefficient	x	0.0000	0.0000
Quality (ISO)	Q	6.0000	6.0000
Center distance (mm)	a	86.9734	
Tooth thickness tolerance	-	DIN 3967 cd25	DIN 3967 cd25
Addendum coefficient	h^*_aP	1.0000	1.0000
Dedendum coefficient	h^*_fP	1.2500	1.2500
Root radius coefficient	ρ^*_fP	0.3800	0.3800
Tip form height coefficient	h^*_{FaP}	0.9600	0.9600
Ramp angle (deg)	α_{KP}	50.0000	50.0000
Protuberance height coefficient	h^*_{prP}	0.0000	0.0000
Protuberance angle (deg)	α_{prP}	0.0000	0.0000
Tip alteration (mm)	k_{m_n}	0.0000	0.0000
Tip radius coefficient (for topping)	ρ^*_{aP}	0.0000	0.0000
Gear material	-	DIN 18CrNiMo7 ($\sigma_{Hlim} = 430$ Mpa)	DIN 18CrNiMo7 ($\sigma_{Hlim} = 430$ Mpa)
Young's modulus (N/mm ²)	E	2.0600×10^5	2.0600×10^5
Poisson's ratio	ν	0.3000	0.3000
Density (kg/m ³)	ρ	7830.0000	7830.0000
Coeff. of thermal expansion (10 ⁻⁸ /°C)	α_{coeff}	11.5000	11.5000
Surface hardness	HRC	61.0000	61.0000
Core hardness	HBW	325.0000	325.0000
Tensile strength (N/mm ²)	σ_B	1200.0000	1200.0000
Yield strength (N/mm ²)	σ_s	850.0000	850.0000
Specific heat capacity (J/kg.K)	c_M	485.0000	485.0000
Specific heat conductivity (W/m.K)	λ_M	50.0000	50.0000
Mean roughness height, root (um)	R_{ZF}	20.0000	20.0000
Mean roughness height, flank (um)	R_{ZH}	4.8000	4.8000
Roughness average value, root (um)	R_{AF}	3.0000	3.0000
Roughness average value, flank (um)	R_{AH}	0.6000	0.6000

	Symbol	Pair1 Gear1	Pair1 Gear2
Fatigue strength for bending stress (N/mm ²)	σ_{Flim}	430.0000	430.0000
Fatigue strength for contact stress (N/mm ²)	σ_{Hlim}	1500.0000	1500.0000

2.2. Calculated geometry - Gear

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Gear ratio	GR	2.2000	
Transverse module (mm)	m_t	2.7179	
Working normal pressure angle (deg)	α_{wn}	20.0000	
Transverse pressure angle (deg)	α_t	21.5885	
Working transverse pressure angle (deg)	α_{wt}	21.5885	
Working transverse pressure angle, e/i (deg)	$\alpha_{wt}, e/i$	21.6068 / 21.5702	
Base helix angle (deg)	β_b	21.6340	
Helix angle at operating pitch circle (deg)	β_w	23.1000	
Radial backlash without tolerance (mm)	j_{r0}	0.0000	
Cumferential backlash without tolerance (mm)	j_{t0}	0.0000	
Min/max circumferential backlash by flank tolerance for each gear (mm)	$j_t, e/i$	0.0761 / 0.1196	0.0761 / 0.1196
Min/max normal backlash by flank tolerance for each gear (mm)	$j_n, e/i$	0.0658 / 0.1034	0.0658 / 0.1034
Cumferential backlash, e/i (mm)	$j_{tw}, e/i$	0.1435 / 0.2479	
Normal backlash, e/i (mm)	$j_{nw}, e/i$	0.1240 / 0.2143	
Angluar backlash, min (deg)	j_θ, min	0.3025	0.1375
Angluar backlash, max (deg)	j_θ, max	0.5226	0.2375
Effective facewidth (mm)	b_{eff}	40.0000	
Sum of the profile shift coefficients	Sum_x	0.0000	
Generating profile shift coefficient	$x_E, e/i$	-0.0385 / -0.0604	-0.0385 / -0.0604
Prevent undercut profile shift coefficient	x_{Emin}	-0.4718	-2.2380
Max. profile shift coeff. (min top land)	x_{Emax}	1.1780	2.0191
Reference center distance, $x_1=x_2=0$ (mm)	a_d	86.9734	
Working center distance (mm)	a_w	86.9734	
Reference diameter (mm)	d	54.3584	119.5884
Operating pitch diameter (mm)	d_w	54.3584	119.5884
Operating pitch diameter with tolerance (mm)	$d_w, e/i$	54.3652 / 54.3515	119.6035 / 119.5733
27		59.3584	124.5884
Tip diameter with tolerance (mm)	$d_a, e/i$	59.3584 / 59.2584	124.5884 / 124.4884

	Symbol	Pair1 Gear1	Pair1 Gear2
Root diameter (mm)	d _f	48.1084	113.3384
Root diameter with tolerance (mm)	d _f , e/i	47.9160 / 47.8062	113.1461 / 113.0362
Tip form diameter (mm)	d _{Fa}	59.1584	124.3884
Tip form diameter with tolerance (mm)	d _{Fa} , e/i	58.9413 / 58.8176	124.1833 / 124.0662
Root form diameter (mm)	d _{Ff}	50.9502	115.2831
Root form diameter with tolerance (mm)	d _{Ff} , e/i	50.8871 / 50.8534	115.1463 / 115.0692
Active tip diameter (EAP) (mm)	d _{Na}	59.1584	124.3884
Active tip diameter with tolerance (mm)	d _{Na} , e/i	58.9413 / 58.8176	124.1833 / 124.0662
Active root diameter (SAP) (mm)	d _{Nf}	51.2155	116.0676
Active root diameter with tolerance (mm)	d _{Nf} , e/i	51.3475 / 51.2815	116.2762 / 116.1713
Mean diameter for mass calculation (mm)	d _m	53.5847	118.8148
Gear mass (kg)	m	0.4312	1.1647
Gear inertia (kg.m ²)	J	0.0000	0.0000
Nominal tip clearance (mm)	c	0.6250	0.6250
Effective tip clearance, e/i (mm)	c, e/i	0.8371 / 0.7102	0.8371 / 0.7102
Min. margin of involute contact, (d _{Nf_i} - d _{Ff_e}) / 2 (mm)	c _{Nff}	0.1972	0.5125

2.3. Calculated geometry - Tooth

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Tooth depth (mm)	h	5.6250	5.6250
Tooth depth, e/i (mm)	h, e/i	5.7761 / 5.6712	5.7761 / 5.6712
Working depth (mm)	h _w	5.0000	
Addendum (mm)	h _a	2.5000	2.5000
Addendum, e/i (mm)	h _a , e/i	2.5000 / 2.4500	2.5000 / 2.4500
Dedendum (mm)	h _f	3.1250	3.1250
Dedendum, e/i (mm)	h _f , e/i	3.2212 / 3.2761	3.2212 / 3.2761
Number of virtual gear teeth	z _n	25.1636	55.3600
Tooth thickness coeff.	s _{nc}	1.5708	1.5708
Transverse tooth thickness (mm)	s _t	4.2693	4.2693
Normal tooth thickness (mm)	s _n	3.9270	3.9270
Normal tooth thickness at d _a (mm)	s _{an}	1.8090	1.9569
Normal tooth thickness at d _a , e/i (mm)	s _{an} , e/i	1.7869 / 1.6906	1.9292 / 1.8430
Normal tooth thickness at d _{Fa} (mm)	s _{Fan}	1.9148	2.0462
Normal tooth thickness at d _{Fa} , e/i (mm)	s _{Fan} , e/i	2.0166 / 1.9101	2.1161 / 2.0234
Normal space width at d _f (mm)	e _{fn}	0.0000	2.0268
Normal space width at d _f , e/i (mm)	e _{fn} , e/i	0.0000 / 0.0000	2.0563 / 2.0738

	Symbol	Pair1 Gear1	Pair1 Gear2
Normal pitch (mm)	p_n	7.8540	
Transverse pitch (mm)	p_t	8.5386	
Normal base pitch (mm)	p_bn	7.3803	
Transverse base pitch (mm)	p_bt	7.9396	
Axial pitch (mm)	p_x	20.0185	
Lead (mm)	p_z	400.3690	880.8119

2.4. Calculated geometry - Mesh

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Length of the path of contact (A to E) (mm)	g_α	11.2399	
Length of path of contact, e/i (mm)	g_α, e/i	10.8301 / 10.5181	
Length of approach POC of pinion (A to C) (mm)	g_f1	5.8705	
Length of approach POC of pinion, e/i (mm)	g_f1, e/i	5.6614 / 5.6203	
Length of recess POC of pinion (C to E) (mm)	g_a1	5.3694	
Length of recess POC of pinion, e/i (mm)	g_a1, e/i	5.1687 / 5.1501	
Length T1 - A (mm)	T1A	4.1298	
T1A tolerance (mm)	e/m/i	4.5210 / 4.4253 / 4.3295	
Length T1 - B (mm)	T1B	7.4300	
T1B tolerance (mm)	e/m/i	7.2201 / 7.1598 / 7.0995	
Length T1 - C (mm)	T1C	10.0003	
T1C tolerance (mm)	e/m/i	10.0096 / 10.0003 / 9.9909	
Length T1 - D (mm)	T1D	12.0694	
T1D tolerance (mm)	e/m/i	12.4606 / 12.3649 / 12.2691	
Length T1 - E (mm)	T1E	15.3696	
T1E tolerance (mm)	e/m/i	15.1597 / 15.0994 / 15.0391	
Length T2 - A (mm)	T2A	27.8711	
T2A tolerance (mm)	e/m/i	27.6414 / 27.5756 / 27.5097	
Length T2 - B (mm)	T2B	24.5708	
T2B tolerance (mm)	e/m/i	24.9312 / 24.8411 / 24.7509	
Length T2 - C (mm)	T2C	22.0006	
T2C tolerance (mm)	e/m/i	22.0211 / 22.0006 / 21.9800	
Length T2 - D (mm)	T2D	19.9315	
T2D tolerance (mm)	e/m/i	19.7018 / 19.6360 / 19.5701	
Length T2 - E (mm)	T2E	16.6312	
T2E tolerance (mm)	e/m/i	16.9916 / 16.9015 / 16.8113	
Length T1 - T2 (mm)	T1T2	32.0009	

	Symbol	Pair1 Gear1	Pair1 Gear2
T1T2 tolerance (mm)	e/m/i	32.0307 / 32.0009 / 31.9710	
Diameter of single contact point A (mm)	d_A	51.2155	124.3884
Diameter of single contact point B (mm)	d_B	52.6843	121.5738
Diameter of single contact point C (mm)	d_C	54.3584	119.5884
Diameter of single contact point D (mm)	d_D	56.0133	118.1285
Diameter of single contact point E (mm)	d_E	59.1584	116.0676
Transverse contact ratio	ϵ_{α}	1.4157	
Overlap ratio	ϵ_{β}	1.9982	
Total contact ratio	ϵ_{γ}	3.4138	
Total contact ratio with tolerance	e/m/i	3.362 / 3.343 / 3.323	
Specific sliding at Tip	ζ_a	0.5081	0.6740
Specific sliding at Root	ζ_f	-2.0677	-1.0331

2.5. Calculated geometry - Accuracy

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Quality Standard	-	ISO 1328: 2013	ISO 1328: 2013
Single pitch deviation (μm)	f_pt	8.5000	8.5000
Base circle pitch deviation (μm)	f_pb	7.9000	7.9000
Sector pitch deviation over k/8 pitches (μm)	F_pk/8T	15.0000	19.0000
Profile form deviation (μm)	f_fa	9.0000	9.0000
Profile slope deviation (μm)	f_Ha	7.0000	7.0000
Total profile deviation (μm)	F_α	11.0000	11.0000
Helix form deviation (μm)	f_fb	10.0000	11.0000
Helix slope deviation (μm)	f_Hβ	9.5000	9.5000
Total helix deviation (μm)	F_β	14.0000	15.0000
Cumulative pith deviation, total (μm)	F_p	25.0000	28.0000
Runout (μm)	F_r	22.0000	25.0000
Single flank composite, tooth-to-tooth (μm)	f_isT	8.5000	8.5000
Single flank composite, total (μm)	F_isT	33.5000	36.5000
Tooth-to-tooth radial composite deviation (μm)	f_idT	9.5000	9.5000
Total radial composite deviation (μm)	F_idT	31.0000	31.0000

2.6. Calculated geometry - Measurement

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Number of teeth spanned	k	3.0000	7.0000
Base tangent length (mm)	W_k	19.3391	49.9263
Base tangent length with tolerance (mm)	W_k, e/i	19.2733 / 19.2357	49.8606 / 49.8230
Tolerance of W_k (mm)	ΔW_k , e/i	-0.0658 / -0.1034	-0.0658 / -0.1034
Diameter of contact point (mm)	d_M	53.6205	120.4651
Theoretical diameter of ball/pin (mm)	D_M.theory	4.2769	4.2217
Effective Diameter of ball/pin (mm)	D_M.eff	4.5000	4.2500
Radial single ball measurement (mm)	M_rk	30.4937	62.7059
Radial single ball measurement with tolerance (mm)	M_rk, e/i	30.4140 / 30.3680	62.6164 / 62.5650
Diametral two ball measurement (mm)	M_dk	60.9874	125.4117
Diametral two ball measurement with tolerance (mm)	M_dk, e/i	60.8279 / 60.7360	125.2328 / 125.1299
Tolerance of M_dk (mm)	ΔM_{dk}	-0.1594 / -0.2513	-0.1789 / -0.2818

3. Detail rating

3.1. General

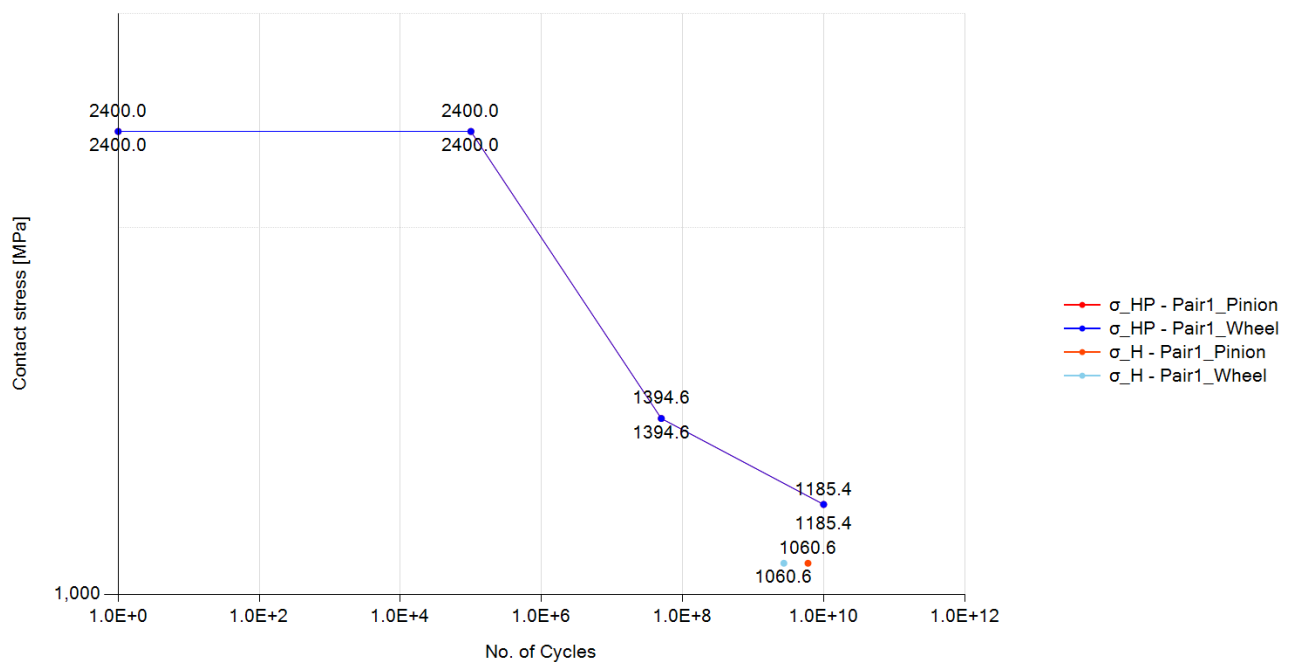
Load case 1, Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Rotational speed (rpm)	ω	10000.0000	4545.4550
Nominal tangential load at d (N)	F_t	5270.1890	
Nominal axial load at d (N)	F_a	2247.9260	
Nominal radial load at d (N)	F_r	2085.3960	
Nominal normal load at d (N)	F_n	6097.2900	
Nominal tangential load per unit facewidth (N/mm)	w	131.7550	
Circumferential speed at d (m/s)	v	28.4620	
Sliding velocity at Tip dia. (m/s)	v_g	8.1790	8.9420
Nominal tangential load at d_w (N)	F_{tw}	5270.1890	
Nominal axial load at d_w (N)	F_{aw}	2247.9260	
Nominal radial load at d_w (N)	F_{rw}	2085.3960	
Circumferential speed at d_w (m/s)	v_w	28.4620	
Running-in value (μm)	y_p	0.5930	
Running-in value (μm)	y_f	0.6750	
Effective single pitch & profile deviation (μm)	f_{pb_eff}	7.3080	
Effective single pitch & profile deviation (μm)	f_{fa_eff}	8.3250	
Correction factor	C_M	0.8000	
Gear blank factor	C_R	0.8980	
Basic rack factor	C_B	0.9750	
Material factor	E/E_{st}	1.0000	
Theoretical single stiffness (N/mm/ μm)	c'_{th}	17.2210	
Single tooth stiffness (N/mm/ μm)	c'	11.0950	
Mesh stiffness (for K_v , $KH\alpha$, $KF\alpha$), (N/mm/ μm)	$c_{y\alpha}$	14.5540	
Mesh stiffness (for $KH\beta$, $KF\beta$), (N/mm/ μm)	$c_{y\beta}$	12.3710	
Mean diameter for K_v (mm)	$d_m (K_V)$	53.7330	118.9630
Moment of inertia for K_v ($\text{kg}\cdot\text{mm}^2$)	$J (K_V)$	247.5610	4286.0140
Moment of gear inertia per unit facewidth ($\text{kg}\cdot\text{mm}$)	J^*	6.1890	107.1500
Relative gear mass per unit facewidth (kg/mm)	m^*	0.0097	0.0347
Relative gear mass per unit facewidth (kg/mm)	m_{red}	0.0076	
Resonance running speed (rpm)	n_{E1}	20931.5930	
Resonance ratio	N	0.4780	
Resonance ratio in main resonance ratio	N_s	0.8500	
Resonance range	-	subcritical range	
Dynamic factor	K_V	1.1490	

	Symbol	Pair1 Gear1	Pair1 Gear2
Face load factor - flank	$K_{H\beta}$	1.3000	
Face load factor - root	$K_{F\beta}$	1.2530	
Transverse load factor - flank	$K_{H\alpha}$	1.1370	
Transverse load factor - root	$K_{F\alpha}$	1.1370	
Number of load cycles	N_L	6.0000×10^9	2.7270×10^9

3.2. Pitting

S-N curve (Pitting), Load case 1, Meshing pair: Gear1 - Gear2



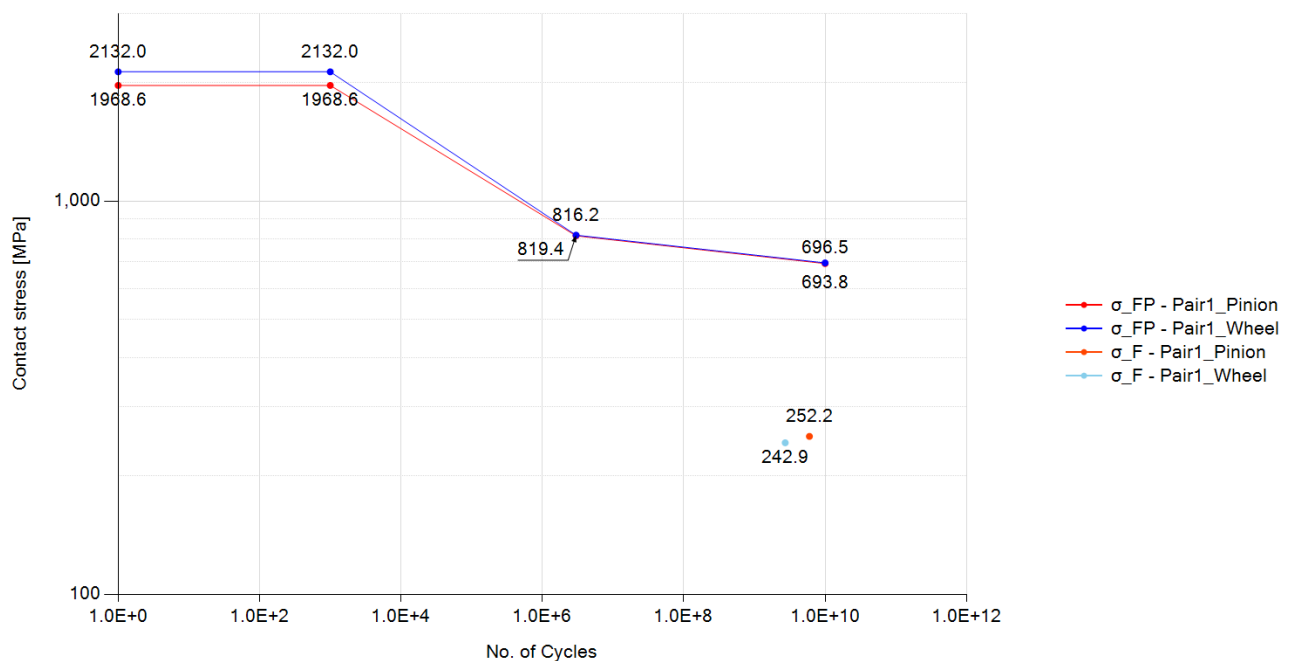
Load case 1, Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Rating Method	-	ISO 6336-2:2019	
Contact ratio factor	Z_ϵ	0.8400	
Zone factor	Z_H	2.3310	
Elasticity factor	Z_E	189.8120	
Young's modulus for tooth flank (N/mm ²)	E	2.0600×10^5	2.0600×10^5
Helix angle factor	Z_β	1.0430	
Single pair tooth contact factor (pinion)	Z_B	1.0000	
Single pair tooth contact factor (wheel)	Z_D	1.0000	
Nominal contact stress at the pitch point (N/mm ²)	σ_{H0}	728.0530	
Contact stress (N/mm ²)	σ_H	1060.6240	1060.6240
Life factor	Z_{NT}	0.8630	0.8850

	Symbol	Pair1 Gear1	Pair1 Gear2
Lubricant factor	Z_L	0.9460	0.9460
Velocity factor	Z_V	1.0310	1.0310
Roughness factor	Z_R	0.9540	0.9540
Work hardening factor	Z_W	1.0000	1.0000
Size factor	Z_X	1.0000	1.0000
Permissible contact stress - static (N/mm ²)	σ_{HP_static}	2400.0000	2400.0000
Permissible contact stress - mid point (N/mm ²)	σ_{HP_mid}	0.0000	0.0000
Permissible contact stress - reference (N/mm ²)	σ_{HP_ref}	1394.6120	1394.6120
Permissible contact stress - long (N/mm ²)	σ_{HP_long}	1185.4200	1185.4200
Permissible contact stress (N/mm ²)	σ_{HP}	1204.1410	1233.6180
Safety factor for Contact	S_H	1.1350	1.1630
Contact damage (%)	-	1.5960	0.7260
Cycle to failure	-	3.7580×10^{11}	3.7580×10^{11}

3.3. Bending

S-N curve (Bending), Load case 1, Meshing pair: Gear1 - Gear2



Load case 1, Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Rating Method	-	ISO 6336-3:2019	
Tooth form factor	Y_F	1.2470	1.0960
Stress correction factor	Y_S	1.8080	1.9810

	Symbol	Pair1 Gear1	Pair1 Gear2
Working angle (deg)	α_{Fen}	18.6070	19.8530
Bending moment arm (mm)	h_{Fe}	2.6050	2.7520
Tooth thickness at root (mm)	s_{Fn}	4.9690	5.4270
Tooth root radius (mm)	ρ_F	1.4530	1.3060
Helix angle factor	Y_{β}	1.0380	
Deep tooth factor	Y_{DT}	1.0000	
Rim thickness factor	Y_B	1.0000	1.0000
Nominal tooth root stress (N/mm ²)	σ_{F0}	123.2860	118.7630
Tooth root stress (N/mm ²)	σ_F	252.1700	242.9180
Life factor	Y_{NT}	0.8590	0.8720
Relative notch sensitive factor	$Y_{\delta relT}$	0.9920	0.9960
Surface factor	Y_{RrelT}	0.9570	0.9570
Size factor	Y_X	1.0000	1.0000
Stress correction factor	Y_{ST}	2.0000	
Mean stress influence factor	Y_M	1.0000	1.0000
Technical factor	Y_T	1.0000	1.0000
Permissible tooth root stress - static (N/mm ²)	σ_{FP_static}	1968.6430	2132.0060
Permissible tooth root stress - reference (N/mm ²)	σ_{FP_ref}	816.1990	819.3920
Permissible tooth root stress - long (N/mm ²)	σ_{FP_long}	693.7690	696.4830
Permissible tooth root stress (N/mm ²)	σ_{FP}	700.9060	714.8520
Safety factor for Bending	S_F	2.7790	2.9430
Bending damage (%)	-	0.0000	0.0000
Cycle to failure	-	4.4050×10^{24}	3.4590×10^{25}

3.4. Scuffing

Load case 1, Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Lubrication type	-	Spray lubrication	
Lubricant temperature (deg)	θ_{oil}	80.0000	
Lubricant viscosity (mPa.s)	η_{oil}	13.1510	
Load stage according to FZG A/8, 3/90	S_{FZG}	12.0000	
Gear density (kg/m ³)	ρ_M	7830.0000	7830.0000
Gear specific heat (J/kg.K)	c_M	485.0000	485.0000
Gear heat conductivity (N/s.K)	λ_M	50.0000	50.0000
Gear roughness (μm)	R_a	0.6000	0.6000
Number of mating gears	n_p	1.0000	
Effective facewidth (mm)	b_{eff}	34.0000	

	Symbol	Pair1 Gear1	Pair1 Gear2
Relevant tip relief (μm) (Min. value = 2 μm)	C_a	2.0000	2.0000
Transverse unit load (N/mm)	w_Bt	279.6170	
Lubrication system factor	X_S	1.2000	
Multiple meshing factor	X_mp	1.0000	
Structural factor	X_W	1.0000	1.0000
Structural factor (total)	X_Wtot	1.0000	
Thermal contact factor	B_M	13.7800	13.7800
Multiple path factor	K_mp	1.0000	
Optimal tip relief (μm)	C_eff	11.3160	
Angle factor	X_αβ	0.9660	
Lubricant factor	X_L	0.8790	
Roughness factor	X_R	0.7750	
Mean coefficient of friction	μ_m	0.0470	
Maximum flash temperature (deg)	θ_fl_max	110.3470	
Mean flash temperature (deg)	θ_fl_mean	35.7670	
Total mass temperature (deg)	θ_M	100.1720	
Scuffing temperature (deg)	θ_S	364.8370	
Maximum contact temperature (deg)	θ_Bmax	210.5190	
Safety factor for flash temperature	S_B	2.1820	

3.5. Micro-pitting

Load case 1

	Symbol	Gear1 - Gear2
minimum lubricant film thickness (μm)	h_Ymin	0.0905
minimum specific lubricant film thickness in the contact area	λ_GFmin	0.1508
permissible specific lubricant film thickness	λ_GFP	0.2111
safety factor against micropitting	S_λ	0.7144
transverse tangential load at reference cylinder per mesh (N)	F_t	5270.1890
transverse load in plane of action (N)	F_bt	5667.7830
reduced modulus of elasticity (N/mm ²)	E_r	2.2637×10 ⁵
thermal contact coefficient of pinion (N/(ms ^{0.5} *K))	B_M1	13779.6040
thermal contact coefficient of wheel (N/(ms ^{0.5} *K))	B_M2	13779.6040
failure load stage according to FVA 54/7 (90°)	-	0.0000
effective arithmetic mean roughness value (μm)	R_a	0.6000
roughness factor	X_R	1.1740
lubricant factor	X_L	1.0000
lubrication factor	X_S	1.2000

	Symbol	Gear1 - Gear2
elasticity factor (N/mm ²) ^{0.5}	Z_E	189.8120
helical load factor	K_By	1.3000
load losses factor	H_v	0.1450
mean coefficient of friction	μ_m	0.0560
profile position at min. lubricant film thickness	-	0.0000
tip relief factor	X_Ca	1.0000
load sharing factor	X_A	0.9180
buttressing factor	X_but,A	1.3000
effective tip relief (μm)	C_eff	0.0000
Applied tip relief in calculation (μm)	C_a	0.0000
sum of tangential velocities at point A (m/s)	v_Σ,A	17.5910
sliding velocity at point A (m/s)	v_g,A	-8.9420
tangential velocity on pinion at point A (m/s)	v_r1,A	4.3250
tangential velocity on wheel at point A (m/s)	v_r2,A	13.2670
oil inlet/sump temperature (°C)	θ_oil	80.0000
bulk temperature (°C)	θ_M	104.0130
flash temperature at point A (°C)	θ_fl,A	133.7340
contact temperature at point A (°C)	θ_B,A	237.7470
pressure-viscosity coefficient at 38 °C (m ² /N)	α_38	1.8390×10 ⁻⁸
pressure-viscosity coefficient at bulk temperature (m ² /N)	α_θM	1.3050×10 ⁻⁸
pressure-viscosity coefficient at point A contact temperature (m ² /N)	α_θB,A	6.4580×10 ⁻⁹
kinematic viscosity at 40 °C (mm ² /s)	ν_40	68.0000
kinematic viscosity at 100 °C (mm ² /s)	ν_100	8.8000
kinematic viscosity at point A contact temperature (mm ² /s)	ν_θB,A	1.3690
dynamic viscosity at 38 °C (Ns/m ²)	η_38	0.0650
dynamic viscosity at bulk temperature (Ns/m ²)	η_θM	0.0070
dynamic viscosity at oil inlet/sump temperature (Ns/m ²)	η_θoil	0.0120
dynamic viscosity at point A contact temperature (Ns/m ²)	η_θB,A	0.0010
normal radius of relative curvature at point A (mm)	ρ_n,A	3.8690
material parameter	G_M	2953.8490
local velocity parameter at point A	U_A	6.6300×10 ⁻¹¹
local load parameter at point A	W_A	0.0003
local sliding parameter at point A	S_GF,A	0.0750
local nominal Hertzian contact stress at point A (Method B) (N/mm ²)	p_H,A	1100.7020
local Hertzian contact stress at point A including the load factors K (Method B) (N/mm ²)	p_dyn,A	1603.4970

4. Power loss

4.1. Mesh loss

Load case 1

	Symbol	Gear1 - Gear2
Rating Method	-	ISO/TR 14179-2:2001
Input Power (kW)	P	150.0000
Gear ratio	u	2.2000
Tangential speed at working pitch dia. (m/s)	v_{wt}	28.4620
Tangential force at pitch dia. (N)	F_t	5270.1890
Tooth normal force, transverse section (N)	F_{bt}	5667.7833
Arithmetic average roughness of pinion	Ra1	0.6000
Arithmetic average roughness of wheel	Ra2	0.6000
Roughness factor	X_R	0.6000
Sum velocity (m/s)	v_{Σ}	20.9445
Equivalent radius of curvature (mm)	ρ	7.3962
Kinematic viscosity of oil at operating temp. (cSt)	η_{oil}	14.8602
Lubricant type	-	Mineral oil base
Lubricant factor	X_L	1.0000
distributed force on tooth surface(N/m)	F/b	150.0000
Mean coefficient of friction of the gear mesh	μ_{mz}	0.0367
Tooth loss factor	H_v	0.1446
Mesh power loss (W)	P_{VZP}	795.6138
Mesh efficiency (%)	η_z	99.4696

4.2. Churning loss

Load case 1

	Symbol	Gear1	Gear2
Rating Method	-	Changenet (2011)	Changenet (2011)
Rotating speed (rpm)	n	10000.0000	4545.4545
Lubricant	-	Oil: ISO-VG 68	Oil: ISO-VG 68
Gear immersed depth (mm)	h_e	27.1792	59.7942
Oil density (kg/m ³)	ρ	885.0000	885.0000
Operating Viscosity (cSt)	ν	14.8602	14.8602
dynamic viscosity (kg/m.s)	μ	0.0000	0.0000
Reynolds number	Re	76612.6568	76612.6568
Froude number	Fr	3038.2582	1381.0265
Centrifugal acceleration (m/s ²)	γ	15303.9212	4112.4275

	Symbol	Gear1	Gear2
Tip diameter of gear (mm)	d_a	59.3584	124.5884
Pitch diameter of gear (mm)	d_p	54.3584	119.5884
Facewidth (mm)	b	40.0000	40.0000
Dimensionless drag torque	C_m	0.0012	0.0029
Immersion area of the gear (m^2)	S_m	0.0109	0.0302
Total oil churning loss (W)	P	136.3410	881.7330
Total oil churning torque (N.m)	T	0.1302	1.8524

4.3. Windage loss

Load case 1

	Symbol	Gear1	Gear2
Rating Method	-	None	None
Windage loss (W)	P_WL	0.0000	0.0000
Windage torque (N.m)	T_WL	0.0000	0.0000